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RESEARCH ARTICLE

Attention to the Tail(s): Global Financial Conditions and Exchange Rate Risks

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Abstract

We document how the entire distribution of exchange rate returns responds to changes in global financial conditions. We measure global financial conditions as the common component of country-specific financial condition indices, computed consistently across a large panel of developed and emerging economies. Using quantile regression, we provide a characterisation and ranking of the *tail* behaviour of a large sample of currencies in response to a tightening of global financial conditions, corroborating (and *quantifying*) some of the prevailing narratives about safe haven and risky currencies. Compared to most standard approaches, our methodology delivers a more nuanced picture of exchange rate behaviour, allowing for example to make probabilistic statements about the likelihood of observing large swings in returns given the prevailing global financial environment. We also identify macroeconomic fundamentals associated with different tail dynamics: currencies of countries with higher interest rates, low levels of international reserves and large fiscal deficits display more marked increases in the likelihood of large losses in response to a tightening of global financial conditions.

Keywords Exchange rates · Tail risks · Financial conditions indices · Global financial cycle · Quantile regression

JEL Classification C22 · F31 · G15

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1 Introduction

Recent years have witnessed a heated debate about the extent and interpretation of the global co-movement of financial variables. Proponents of a so-called global financial cycle, beginning with Rey (2013), argue that the observed cross-country co-movement in asset prices cannot be fully explained by co-movement in real variables alone and therefore must have a finance-specific component to it, such as some measure of global risk aversion. Others, such as Cerutti et al. (2017), argue against the very notion of a 'global financial cycle'.

Within that debate, there is an asset class that stands out from the rest: exchange rates. Being relative prices, the scope for them to co-move at the global level is limited by construction. Moreover, the relationship between exchange rate movements and overall financial conditions in a country is not a priori obvious, as a given exchange rate move can have heterogeneous effects across different sectors. For example, an exchange rate depreciation can 'tighten' access to finance for agents with net balance sheets exposures in foreign currency (as their net worth falls and interest payments become more expensive in domestic currency terms), while 'loosening' access for exporters (as their external demand prospects improve).

Against this backdrop, we study how different exchange rates co-move with global financial conditions. Given that exchange rates typically exhibit 'fat tails' (i.e. non-negligible chances of a large depreciation and/or appreciation), we put a special emphasis on studying such occurrences, which are of great importance to both policy makers concerned about financial stability and investors seeking ways to quantify the value-at-risk of their strategies. In order to do so, and unlike most of the existing literature, we go beyond mean-based approaches and instead study the behaviour of the *entire distribution* of different currencies' returns in the face of changes in global financial conditions.

We begin by constructing a new measure of global financial conditions, a principal component-based index that has two main advantages over existing alternatives. First, it is truly 'global' in nature, drawing from financial asset prices in more than 40 countries. Second, the within-country coverage across asset classes is broad, including information from a range of public and private spreads as well as equity markets. The resulting index closely co-moves with a number of other commonly used proxies such as the VIX or the indices proposed by Miranda-Agrippino and Rey (2015), Bekaert et al. (forthcoming) and Chari et al. (2020), among others, but its ability to fit the tail behaviour of exchange rates is typically somewhat better. Nevertheless, many of our key results still hold when using proxies of global financial conditions previously studied in the literature, and reinforced when such proxies are used jointly with the one we propose.

We then study the co-movement with global financial conditions of a broad set of currencies. Our point of departure are 'market narratives' and earlier contributions (Ranaldo and Soderlind 2010; Habib and Stracca 2012, among others) that typically label currencies as 'safe havens' or 'risky', depending on their performance in the face of a tightening of global financial conditions. For example, the

Japanese yen (JPY) is typically regarded as a 'safe haven' currency as it tends to appreciate on such occasions, while the Australian dollar (AUD) tends to depreciate, and is therefore typically placed in the 'risky' camp. In this paper, we go beyond this narrative characterisation and provide a quantification of the (tail) risks facing particular currencies under different scenarios for global financial conditions.

In order to provide quantitative insights, we exploit several novel possibilities afforded by quantile regression (Koenker and Bassett 1978). Figure 1 shows the typical output of our regressions for the JPY and AUD, with quantile-specific coefficients in blue and standard regression coefficients in black. The two panels show the very different effects of a one standard deviation tightening in global financial conditions on the two currencies: for the JPY, most of the conditional distribution shifts to the right, increasing the chances of an appreciation, while the pattern for the AUD indicates an increased risk of a sharp depreciation. Moreover, the pattern of quantile coefficients highlights that the strongest effects are concentrated in the tails of the two distributions, which tends to hold across our panel of currencies, and is a feature that cannot be uncovered using standard regression.

These estimates allow us to provide interesting insights, such as the shift in the probability of a currency experiencing a depreciation/appreciation of a given magnitude in the face of different scenarios for global financial conditions. So for example, when global financial conditions tighten by one standard deviation, the chance of the AUD experiencing a depreciation of 2.5-5% increases by 5 p.p., while the corresponding chance of a similar appreciation for the JPY increases by 7 p.p. (see Table 4). One practical application is the ability to assess the extent to which some of the large currency moves observed at the height of the COVID-19 crisis were 'warranted' by the concomitant tightening in global financial conditions (see Sect. 3.4). We also propose intuitive metrics for cross-currency comparisons and rankings based on tail behaviour.

We also revisit the existing literature on the macro-financial determinants of currency behaviour (see Fratzscher 2009 and Habib and Stracca 2012, among others) in light of our expanded focus on the tails. Specifically, we conduct portfolio sorting exercises based on several macroeconomic fundamentals, and study the responses of the resulting returns series to a tightening in global financial conditions to identify potential macro-financial risk factors associated with different currencies' tail behaviour. This methodology allows both to introduce a degree of time variation in the exposure of currencies to a given risk factor and to estimate the effect of these risks from the entire cross section of currencies in our sample.

We find that currencies of countries displaying high interest rates, low levels of international reserves, larger fiscal deficits, lower current account balances and smaller net foreign asset positions display a significantly higher likelihood of depreciating in response to a tightening in global financial conditions. However, only the first three of those risk factors are associated with a larger 'crash risk'.[1] This exercise also yields a simple tabular mapping from changes in risk factors to shifts in

[1] Defined as a depreciation larger than 2.5% in our exercise.

currencies' tail risks. For example, a country that experiences an 8.2 p.p. increase in its interest rate differential with respect to the rest of the world should see approximately a 25 p.p. increase in the probability of its currency depreciating by more than 2.5% in the event of a sharp tightening of global financial conditions. For practitioners, our results provide further empirical motivation for a close scrutiny of the identified risk factors when assessing countries' financial stability prospects and the risks of investment strategies.

1.1 Related Literature

Our paper is related to several literature strands. First, and most directly, it is related to papers that study the occurrence of tail events in exchange rate markets. On the negative returns side, there is a large literature that documents the existence of 'crash' or 'disaster' risk in popular FX strategies. Brunnermeier et al. (2009) find that carry trade strategies perform particularly poorly during periods of heightened risk aversion (as proxied by the VIX index), while Menkhoff et al. (2012) show similar results but focusing on periods of high FX volatility. Relatedly, Farhi and Gabaix (2016) and Farhi et al. (2009) study disaster risk embedded in option prices.

In principle, the poor performance of carry trades could be the result of both a sharp depreciation of high-interest-rate currencies and/or a sharp appreciation of low-interest-rate currencies. In that vein, some papers study the dynamics of particular currencies, namely those usually labelled as safe havens, which, according to market narratives, tend to appreciate sharply during periods of high risk aversion. Ranaldo and Soderlind (2010) and Habib and Stracca (2012) study the safe haven property of a series of currencies, and do indeed find robust evidence of substantial appreciation during periods of market stress. Fratzscher (2009) also looks at the dynamics of individual currencies under stress conditions in the context of the global financial crisis.

A common feature of these papers is that their empirical strategies focus on the (conditional) mean returns of currencies or trading strategies. In contrast, our approach allows for a detailed study of the entire distribution of exchange rate returns, including the tails, which are at the core of our analysis. Moreover, we propose a novel way of characterising periods of heightened (global) risk aversion, avoiding popular but imperfect proxies (e.g. the VIX index), or FX-based proxies which can become somewhat circular (e.g. FX volatility). Moreover, we study a large panel of exchange rates, facilitating a direct analysis of particular currencies.

The paper is also part of a recent wave of contributions drawing on insights from quantile regression, originally introduced in economics by Koenker and Bassett (1978), in both macroeconomics and finance. These include Cenedese et al. (2014) for exchange rates, Gaglianone and Lima (2012) for unemployment, Korobilis (2017) for inflation, Crump et al. (2018) for equity returns, Eguren-Martin et al. (2020) and Gelos et al. (2019) for capital flows, among several others. Most closely

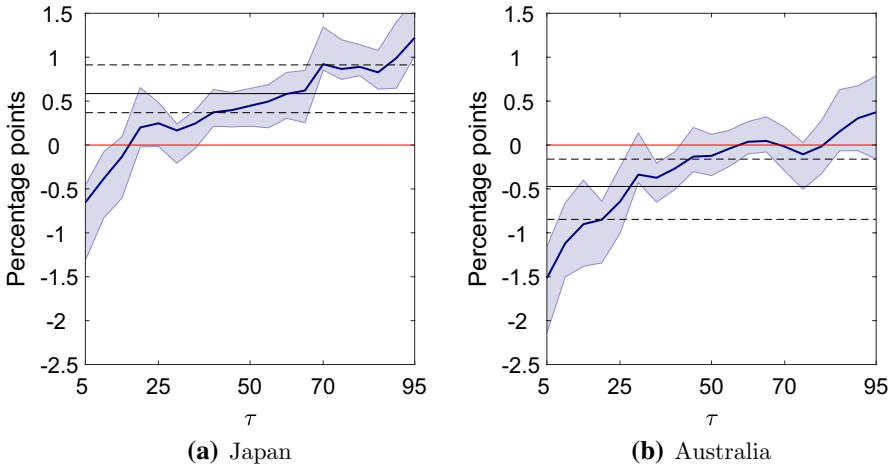


Fig. 1 Impact of global financial conditions on the conditional quantiles of exchange rate returns, JPY and AUD. *Note* The figure shows the coefficients resulting from estimating our baseline specification for the Japanese yen and the Australian dollar (separately). The blue lines plot the values of $\beta_h(\tau)$ across quantiles, while the black lines show OLS estimates of the same specification. Light blue areas are 68% confidence intervals are computed from 1000 overlapping block bootstrap draws. (Color figure online)

related to our study, Adrian et al. (2019) rely on quantile regression to characterise the tails of the GDP growth distribution conditional on domestic financial conditions.[2] We build on similar ideas, but focus instead on the distribution of exchange rate returns conditional on *global* financial conditions, with many parallels to the analysis of capital flows' push factors in Eguren-Martin et al. (2020).

The last strand of literature we draw and build on deals with measurement of financial conditions. We follow Arregui et al. (2018) in constructing country-specific financial condition indices that exploit a broad set of market-based indicators for a large panel of countries, which then allows us to extract a *global* financial conditions index. This exercise is related to earlier attempts to characterise a 'global financial cycle', most notably by Miranda-Agrippino and Rey (2015), but in the finance literature it also overlaps with various proposals to measure global risk aversion and other factors commonly used to price exchange rate rates (see e.g. Menkhoff et al. (2012) and Lustig et al. (2011)).[3] More recent attempts to characterise financial conditions drawing from a range of different financial instruments include Chari et al. (2020) and Bekaert et al. (forthcoming).

The rest of the paper is organised as follows: in Sect. 2, we describe our measure of global financial conditions. In Sect. 3 we discuss quantile regressions of nominal effective exchange rate returns on global financial conditions. In Sect. 4 we introduce a currency portfolio sorting approach based on macroeconomic fundamentals

[2] Relatedly, Adrian et al. (2018) explore the term-structure of this relationship.

[3] See Cerutti et al. (2017) for a contrarian view on the existence of a global financial cycle. Also see Drehmann et al. (2012) for a characterisation of a more medium-term (domestic) financial cycle.

that allows us to identify potential factors associated with currencies' differential tail behaviour. In Sect. 5 we run a series of robustness checks on our results. In Sect. 6 we conclude, while the Appendix includes details about our data and methodology. An online appendix provides additional results and robustness checks.

2 Measuring Global Financial Conditions

The existence of a global factor in financial conditions has been widely debated in economics over recent years.[4] Beginning with Rey (2013), a series of papers have emphasised (and measured) a strong co-movement in financial variables across countries (among others, see Bruno and Shin (2014), Cesa-Bianchi et al. (2018a, 2018b), Ha et al. (2018)). These papers have suggested that this co-movement in financial conditions goes beyond a reflection of co-movement in macroeconomic indicators, and hence is at least partly driven by a specific global factor in financial variables, such as risk appetite. The standard approach has been to measure common variation in a set of asset prices and/or credit quantities, interpreting the result as an indicator of the ease at which finance could be accessed at a given time in a given country (see, for example, Miranda-Agrippino and Rey (2015)).

Existing measures of global financial conditions typically suffer from two shortcomings. First, the breadth of financial series considered tends to be limited, and usually skewed towards equity markets (as, for example, in Miranda-Agrippino and Rey (2015)). Second, the geographical coverage tends to be limited to advanced economies (e.g. Ha et al. (2018)) and, in some cases, a handful of emerging countries. Both of these limitations are due to data availability constraints: it is not straightforward to construct a panel dataset spanning a broad set of financial indicators for a large cross section of countries.

In order to overcome these limitations we follow Arregui et al. (2018) and construct a panel dataset covering a broad set of monthly financial indicators for 43 countries from April 1995 to June 2018. The financial series included are as follows: term, sovereign, interbank and corporate spreads, long-term interest rates, equity returns and volatility and relative market capitalisation of the financial sector.[5] We rely on principal component analysis to extract country-specific summary measures of financial conditions (which correspond to the first principal component of the series considered).[6]

[4] This factor has typically been referred to as 'the global financial cycle'.

[5] A detailed description of the variables used and corresponding data sources can be found in Appendix A.

[6] Note that the resulting first principal component of the series considered is very similar to the common factor obtained when following Arregui et al. (2018) and relying on the method of Koop and Korobilis (2014), which allows for time variation in the parameters and attempt to 'clean' financial conditions from changes that reflect a response to macro-economic news (proxied by industrial production and CPI inflation). This can be interpreted as a result of the relative stability of the parameters and the fact that asset prices tend to react to news about *expected* rather than realised macroeconomic aggregates. Arrigoni et al. (2020) reach a similar conclusion.

Armed with a set country-specific financial condition indices, we then compute a global component simply as their cross-sectional mean.[7]

The share of variance of individual countries' FCIs explained by this global component varies in the cross section, but averages around 35%. It is worth noting that this figure goes well above 50% for several countries, including financial centres such as the US or the UK (see Sect. 1 of the Online Appendix). In what follows we take this series as our measure of global financial conditions.

Figure 2 shows that the evolution of our measure over the last 20 years is broadly in line with prevailing financial market narratives (see e.g. Brunnermeier 2009; Rostagno et al. 2021): for example, the left panel shows that global financial conditions tightened sharply around the collapse of Lehman Brothers in 2008 or the euro area crisis of 2010-2012. The right panel also highlights that our measure exhibits both skewness and kurtosis (fat tails), meaning both that tightenings in global financial conditions have tended to be somewhat larger than loosenings, and that larger outturns have occurred more frequently than implied by a standard Normal distribution. These features are shared by many of the existing proxies of global financial conditions, as shown in Table 1.

Table 2 also shows the correlation of our index with other proxies previously used in the literature to measure both global financial conditions and narrower conditions in FX markets. Our proxy co-moves positively, but far from one-to-one, with other widely used US-centric measures such as the VIX index or TED spreads, and with the estimated factors in Miranda-Agrippino and Rey (2015), Bekaert et al. (forthcoming) and Chari et al. (2020). In the context of this paper, it is also interesting to note that our measure of global financial conditions displays heterogeneous correlations with factors previously used to price exchange rates: it displays a relatively high correlation with the FX volatility factor in Menkhoff et al. (2012), but very low correlations with the dollar and HML factors in Lustig et al. (2011). This is particularly interesting because these factors are computed using the very same exchange rate data that are then priced with them, while our measure does not directly contain any FX data at all.

While our proposed measure arguably represents an improvement over existing proxies of the global financial cycle in terms of both geographical and asset class coverage, many of our results on the link between global financial conditions and exchange rates' tail behaviour do not rely on it and hold through with a number of the proxies listed in Table 2. Section 5.4 elaborates on this point, but also shows that even if the various proxies are typically highly correlated, there can still be significant gains in fit in the tail regions of exchange rate returns from their joint inclusion as explanatory variables.

[7] Taking the first principal component of the indices yields an almost indistinguishably similar series, see Sect. 1 of the Online Appendix.

3 Assessing Exchange Rate Tail Risks with Quantile Regression

As discussed in Sect. 2, asset prices tend to display a high degree of co-movement across countries. However, exchange rates are somewhat special. Being relative prices, the pattern and extent of their co-movement is more constrained than for other assets. This feature is the departing point of our analysis: we want to understand how different exchange rates co-move with changes in global financial conditions, and the underlying country-specific characteristics that are associated with such dynamics.

Our focus is on the entire distribution of exchange rate returns, and in particular on tail events. Specifically, we study how the likelihood of sharp exchange rate movements (in either direction) is affected by global financial conditions. To this end, we rely on quantile regression, originally introduced in economics by Koenker and Bassett (1978) and recently popularised by Adrian et al. (2019), among others. Unlike standard regression, which provides an estimate of the conditional mean of a variable of interest given a set of explanatory variables, quantile regression allows to model the entire conditional distribution of a dependent variable given a set of covariates. This allows to capture features that are lost when only focussing on the average response, as already highlighted in our discussion of Fig. 1 in the Introduction.

Given our special emphasis on measuring tail risks, it is worth highlighting that quantile regression also offers at least one important advantage relative to probit/logit-type frameworks, for example as applied to capital flows surges and retrenchments by Forbes and Warnock (2012) (see also the discussion in Eguren-Martin et al. 2020). In such frameworks, the tails events need to be defined *ex-ante* with reference to *ad-hoc* thresholds. In contrast, quantile regression allows to directly map risk factors to any part of the conditional distribution of a variable of interest. The same logic applies to the analysis of exchange rates.

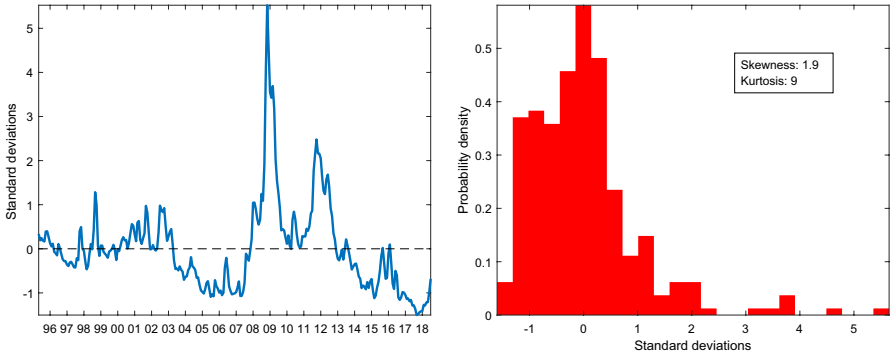
3.1 Specification

Following the (limited) existing literature applying quantile regression to exchange rates (see, for example, Cenedese et al. 2014), our baseline exercise studies the effect of global financial conditions on the distribution of exchange rate returns. We specify a linear model for their conditional quantiles as follows:

$$Q_{\Delta FX_{t+h}}(\tau|X_t) = \alpha_h(\tau) + \beta_h(\tau)GFC_t \tag{1}$$

where ΔFX_{t+h} is the monthly log change in the nominal effective exchange rate h months ahead and GFC_t is our measure of global financial conditions.[8] Q thus denotes the $\tau-$th quantile of the distribution of ΔFX_{t+h} conditional on a set of

[8] The convention we adopt here is that positive FX changes represent an appreciation, and negative changes a depreciation. This is done in order to facilitate comparison with the next section, which looks at returns.



(a) Time series **(b)** Histogram

Fig. 2 Global Financial Conditions Index, 1995-2018. *Note* The figure shows the time series of our Global Financial Conditions Index. The index is standardised, so has zero mean and unit variance over the whole sample. Higher values signal tighter financial conditions

Table 1 Higher moments of selected proxies of global financial conditions

	GFCI	VIX	MAR	RORO	BEX
Skewness	1.9	1.9	0.6	2.0	3.8
Kurtosis	9.0	9.0	3.4	8.5	21.2

The table show the skewness and kurtosis of the series over their respective samples. Positive skewness indicates that the mean of the distribution is larger than its mode, i.e. the distribution is skewed to the right. A kurtosis larger than 3 indicates 'fatter tails' than a standard Normal benchmark. GFCI refers to our Global Financial Conditions Index, VIX to the VIX index, MAR to the index from Miranda-Agrippino and Rey (2015), RORO to the (cumulated) index from Chari et al. (2020) and BEX to the index in Bekaert et al. (forthcoming)

covariates X_t (in this case, GFC_t and a constant). Appendix B provides technical details.

Equation (1) is very parsimonious, but we have found that it reliably captures the impact of GFC_t on the conditional quantiles of exchange rate returns. Considering alternative functional forms and additional regressors does not results in significant changes to the estimates of $\beta_h(\tau)$ from the baseline linear specification (while it can of course increase in-sample fit, see Table 6). For example, we have tried adding a quadratic term in order to capture potential nonlinearities, and experimented with adding (i) lags of ΔFX_{t+h} , (ii) additional regressors that aim at capturing the factor structure usually found in exchange rates (specifically the 'dollar' and 'carry' factors in Lustig et al. 2011 and (iii) alternative/additional proxies of global financial conditions found in the literature (for example Miranda-Agrippino and Rey 2015; Bekaert et al. (forthcoming); Chari et al. 2020). Some of these exercises are discussed in more detail in Sect. 5.

Table 2 Correlation between Global Financial Conditions Index and selected variables

	GFCI	VIX	TED	S&P	FXVOL	MAR	MARII	HML	DOL	DOLII	BEX	RORO
GFCI	1.00	0.81	0.49	0.31	0.68	0.71	0.62	0.04	0.02	0.22	0.83	0.73
VIX	0.81	1.00	0.50	0.36	0.73	0.61	0.54	–0.12	–0.28	0.19	0.88	0.78
TED	0.49	0.50	1.00	–0.07	0.45	0.27	–0.18	–0.13	–0.14	0.17	0.46	0.25
S&P	0.31	0.36	–0.07	1.00	0.09	–0.04	0.36	–0.72	–0.63	0.16	0.22	0.63
FXVOL	0.68	0.73	0.45	0.09	1.00	0.59	0.45	0.13	–0.03	0.25	0.74	0.70
MAR	0.71	0.61	0.27	–0.04	0.59	1.00	0.74	0.37	0.31	0.12	0.80	0.84
MARII	0.62	0.54	–0.18	0.36	0.45	0.74	1.00	0.24	0.19	0.07	0.64	0.90
HML	0.04	–0.12	–0.13	–0.72	0.13	0.37	0.24	1.00	0.86	–0.10	0.07	–0.10
DOL	0.02	–0.28	–0.14	–0.63	–0.03	0.31	0.19	0.86	1.00	–0.13	–0.09	–0.11
DOLII	0.22	0.12	0.07	0.19	0.25	0.17	–0.16	–0.10	–0.13	1.00	0.21	0.05
BEX	0.83	0.80	0.64	0.88	0.74	0.46	–0.22	0.07	–0.09	0.21	1.00	0.77
RORO	0.73	0.84	0.90	0.78	0.70	0.25	–0.63	–0.10	–0.11	0.05	0.77	1.00

The table show correlations between series measured over common samples, which vary across pair of indices. GFCI stands for our Global Financial Conditions Index, MAR and MAR II are the 'short' and 'long' factors in Miranda-Agrippino and Rey (2015) respectively, VIX is the VIX index, TED stands for TED spreads, S&P for the (negative of) the S&P500 Index, FXVOL for the volatility factor in Menkhoff et al. (2012) and FX HML and DOL for the HML and dollar factors in Lustig et al. (2011). DOLII is the broad dollar index from the BIS (as proposed in Avdjiev et al. (2019)). BEX is the financial conditions index in Bekaert et al. (forthcoming) and RORO is (the cumulative version of) the index put forward by Chari et al. (2020)

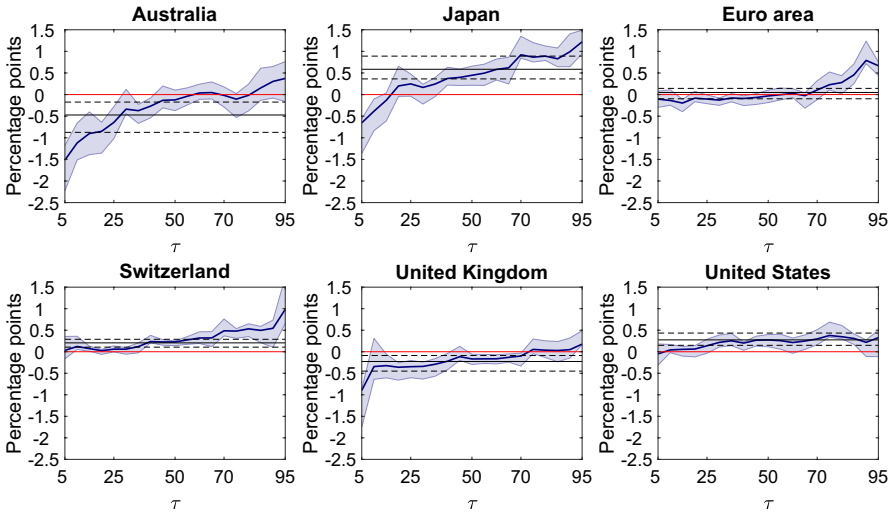


Fig. 3 Impact of global financial conditions on the conditional quantiles of exchange rate returns, selected currencies. *Note* The figure shows the coefficients resulting from estimating our baseline specification for each selected currency (separately). The blue lines plot the values of $\beta_h(\tau)$ across quantiles, while the black lines show OLS estimates of the same specification. Light blue areas are 68% confidence intervals are computed from 1000 overlapping block bootstrap draws. (Color figure online)

We estimate Eq. (1) on a currency-by-currency basis for a panel of 61 countries from April 1995 to June 2018. The full list of currencies can be found in Appendix A. In our baseline, we focus on nominal effective exchange rates to identify idiosyncratic dynamics in exchange rate returns, rather than on potentially US-driven dollar bilaterals.[9] Also, following the existing literature (e.g. Habib and Stracca 2012; Cenedese et al. 2014), we focus on the contemporaneous (monthly) relationship, i.e. on $h = 0$.

Figure 3 reports coefficient estimates of $\beta_0(\tau)$ in the same format as Fig. 1, but now for a set of 'popular' currencies.[10] As previously discussed, the responses of the Japanese yen and the Australian dollar to a tightening of global financial conditions are starkly different, and corroborate their typical labelling as safe-haven and risky, respectively. The Swiss franc and the US dollar are also firmly in the safe-haven camp where market narratives typically place them: across quantiles, their coefficients are either larger than or statistically not different from 0, and increasing for higher quantiles, meaning that when global financial conditions deteriorate, the distribution of their returns shifts and skews to the right, making large appreciations more likely.

[9] See Sect. 5 for a robustness exercise which re-estimates our baseline specification considering US dollar bilateral exchange rates, and another one that considers excess returns instead of plain exchange rate changes.

[10] The full set of coefficient estimates for our panel of currencies is reported in Sect. 2 of the Online Appendix.

On the other hand, our estimates for the UK pound suggest that it tends to behave like a risky currency, as its coefficients are mostly negative, and more so for lower quantiles. The euro is an interesting intermediate case, perhaps owing to the gradual evolution of its underlying economic and political underpinnings over our sample. It exhibits some features of a risky currency, with slightly negative coefficients up to the median, but also clear marks of safe haven status, with large and increasing coefficients in the upper tail.

In the following subsections, we build further on these initial intuitions, and showcase how our regression estimates can be turned into quantitative insights for practitioners and policy-makers.

3.2 Goodness of Fit

Quantile-specific measures of goodness of fit represent a distinctive advantage of quantile regression over more traditional approaches that focus on the conditional mean, and provide an intuitive quantification of the role played by global financial conditions in driving tail behaviour for different currencies. We follow Koenker and Machado (1999) and report quantile-specific $R^1(\tau)$ measures for all currencies. Unlike standard R^2 measures, which quantify the relative success of two models for the conditional mean function, and thus provide a global measure of goodness of fit over the entire conditional distribution, $R^1(\tau)$ measures provide information on the relative local success of two models of a conditional quantile function.

The (adjusted) $R^1(\tau)$ is defined as

$$R^1(\tau) = 1 - \frac{\hat{V}(\tau)}{\tilde{V}(\tau)} \frac{T-1}{T-k} \tag{2}$$

where $\hat{V}(\tau)$ denotes the sum of weighted absolute residuals of model (1) and $\tilde{V}(\tau)$ the sum of weighted absolute residuals of a model consisting only of a constant (which provides an estimate of the unconditional quantile τ).[11] T denotes the length of the sample, and k the number of coefficients in the more complex model. The interpretation is thus analogous to that of standard R^2: $R^1(\tau)$ expresses the improvement in fit, in terms of the relevant criterion function, obtained by adding covariates to the model, with a small penalty for the number of additional coefficients.

Table 3 reports both R^2 and $R^1(\tau)$ measures for the same selection of currencies shown in Fig. 3. The first thing to note is that the overall improvement in fit from including our measure of global financial conditions as a covariate, proxied by the R^2 of a standard OLS regression, varies across countries, and can be rather limited.

On the other hand, as far as $R^1(\tau)$ measures are concerned, a robust pattern seems to hold across countries, namely, that the goodness of fit tends to generally

[11] As explained in Appendix B, $\hat{V}(\tau)$ and $\tilde{V}(\tau)$ are simply the objective functions of the respective quantile regression problems, which take the form of weighted sums of absolute residuals, evaluated at the optimum.

improve in the tails, highlighting the information content of global financial conditions that can be lost by exclusively focusing on the conditional mean of exchange rate returns.[12] It is also worth noting that the improvements tend to be concentrated in one tail, and the largest gains tend to accrue to the most extreme percentiles, so the 95th for the Japanese yen, Swiss franc, US dollar and the euro, and the 5th for the Australian dollar and UK pound.

3.3 Summary Measures of Tail Behaviour

The information conveyed by quantile-specific coefficients estimates can also be summarised visually by studying their effect on fitted probability density functions. In the same spirit as Adrian et al. (2019), who fit skew-*t* distributions to the predictive quantiles of GDP growth, we fit nonparametric density functions to the quantiles of exchange rate returns conditional on different values of global financial conditions.[13] This counterfactual exercise allows both to visualise in an intuitive way the behaviour implied for each currency by our estimates, and to rank currencies based on their tail behaviour. Appendix B provides technical details.

Figure 4 illustrates the changes induced by a one standard deviation increase in global financial conditions on the conditional densities of the same set of currencies analysed before.[14] Focussing again on the JPY and AUD, the right (appreciation) tail of the distribution of the Japanese yen shifts up significantly (increased chances of a sharp appreciation), while the left (depreciation) tail of the distribution of the Australian dollar shifts down (increased chances of a sharp depreciation). Similarly, the exercise implies a 'fattening' of the right tail of the euro, the Swiss franc and the US dollar, and of the left tail of the UK pound. These 'fatter' tails thus not only confirm market narratives but, most importantly, also provide a quantification of the shift in risks.

To compare such heterogeneous tail behaviour across currencies, we compute measures of divergence between each pair of distributions in Fig. 4. In particular, we use a version of the Kullback–Leibler divergence, also known as relative entropy, to quantify the 'shifts' induced in the tail regions by a tightening of global financial conditions.[15] Given a fitted distribution $\hat{g}(x)$ conditional on average global financial conditions and another, $\hat{f}(x)$, conditional on a one standard deviation increase in global financial conditions, we compute downside and upside (relative) entropy outside of the interquartile range of $\hat{g}(x)$ as

[12] This also holds in term of the full panel of currencies, see Sect. 2 of the Online Appendix.

[13] The reason for using a nonparametric distribution is to allow for a more nuanced depiction of tail behaviour than is possible by only fitting a few parameters (four in the case of the skew-*t*) to the estimated conditional quantiles, which is particularly important in our application. A related approach, also based on kernel densities, is followed by Gaglianone and Lima (2012) and Korobilis (2017).

[14] For the full panel see Sect. 2 of the Online Appendix.

[15] This is similar in spirit to the quantification of upside and downside risks in Adrian et al. (2019).

Table 3 Goodness of fit measures, selected currencies

	$R^1(\tau)$					R^2
	0.05	0.25	0.5	0.75	0.95	
Australia	16.2	2.5	–0.2	–0.3	1.3	4.1
Euro area	0.7	–0.1	–0.4	0.2	9.9	–0.2
Japan	2.0	0.3	1.4	5.5	13.8	5.4
Switzerland	–0.3	–0.2	2.4	3.8	6.5	1.9
United Kingdom	7.6	1.9	0.3	–0.3	–0.2	1.8
United States	–0.2	0.0	1.6	1.7	7.2	4.4

The table shows quantile-specific goodness of fit measures $R^1(\tau)$ across currencies, which are computed as shown in Eq. 2, following Koenker and Machado (1999) but adjusting for the number of model parameters. For OLS regressions, the adjusted R^2 is computed in the usual way

$$\mathcal{L}^D = \int_{-\infty}^{\hat{G}^{-1}(0.25)} \log\left(\frac{\hat{f}(x)}{\hat{g}(x)}\right)\hat{f}(x)dx \tag{3}$$

$$\mathcal{L}^U = \int_{\hat{G}^{-1}(0.75)}^{\infty} \log\left(\frac{\hat{f}(x)}{\hat{g}(x)}\right)\hat{f}(x)dx \tag{4}$$

Intuitively, downside and upside entropy measure the additional probability mass assigned to tail events when there is a tightening of global financial conditions. For safe haven currencies, upside entropy should be positive (denoting an increased probability of a large appreciation), whereas for risky currencies, downside entropy should be positive.

Figure 5 shows the results for the same selection of currencies as Fig. 4.[16] The ranking in terms of tail behaviour once again broadly confirms prevailing narratives: typical safe haven currencies such as the Japanese yen and the Swiss franc exhibit high upside entropy but hardly any downside entropy, whereas risky currencies such as the Australian dollar tend to exhibit a higher downside entropy. Both the US dollar and the euro display a similarly high upside entropy ('safe haven'-type behaviour), but it is worth noting that when repeating the exercise for bilateral exchange rates we find an increased likelihood of a sharp euro depreciation vis-à-vis the US dollar (high downside entropy) in the event of a tightening in global financial conditions (see Sect. 5.2).

To provide a more tangible measure, Table 4 also reports changes in appreciation and depreciation probabilities induced by a tightening of global financial conditions; that is, the integral of $\hat{f}(x) - \hat{g}(x)$ over different ranges. Very large swings in returns in either direction (>5%) are never assigned very high probabilities, whereas appreciations or depreciations between 2.5% and 5% tend to be assigned higher chances,

[16] See Sect. 2 of the Online Appendix for the full sample of currencies.

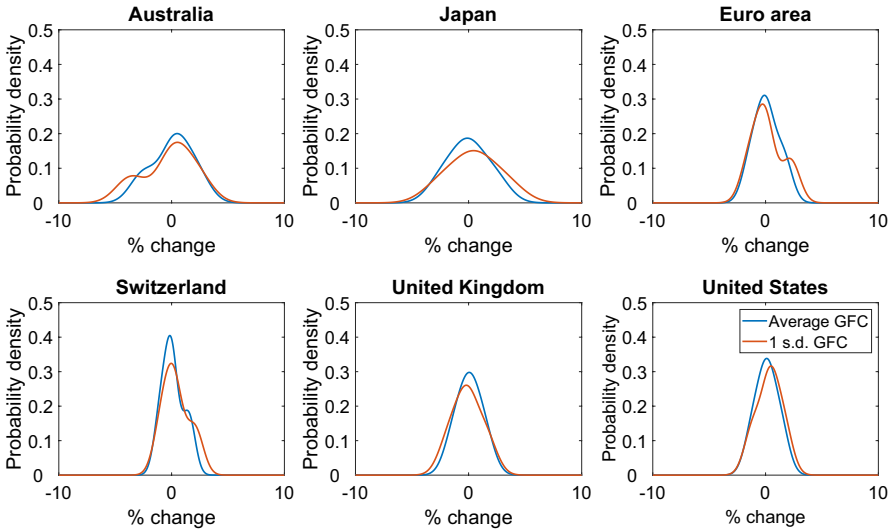


Fig. 4 Impact of a tightening of global financial conditions on the conditional distribution of exchange rate returns. *Note* The panels show, for each currency, nonparametric probability densities fitted to our estimated quantiles conditioning on an average (zero) value of global financial conditions (in blue), and on a one standard deviation increase in global financial conditions (in red). (Color figure online)

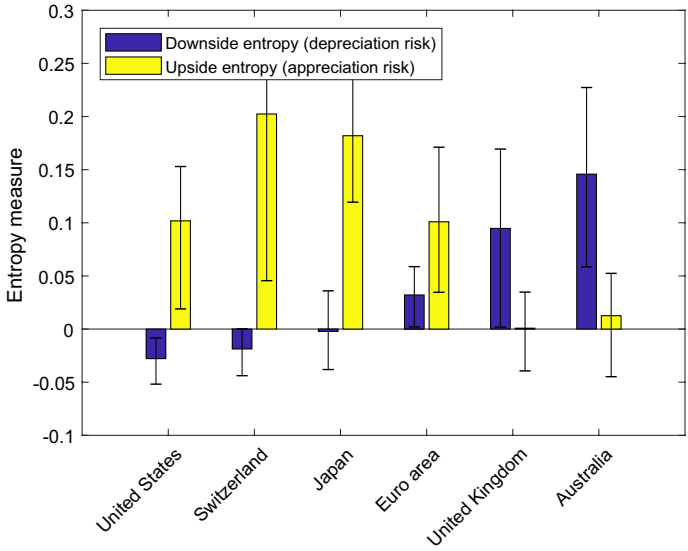


Fig. 5 Downside and upside entropy measures of conditional exchange rate returns, selected currencies. *Note* The figure shows downside and upside entropy measures for a series of currencies, as defined in Eqs. (3) and (4). A positive entropy denotes a higher probability mass assigned to outturns in the corresponding tail by the distribution conditional on tighter global financial conditions than the distribution conditional on average global financial conditions. Black lines quantify 68% confidence intervals to these measures, computed from 1000 overlapping block bootstrap draws. (Color figure online)

Table 4 Changes in appreciation and depreciation probabilities due to a tightening of global financial conditions, selected currencies

	Depreciation probability Δ			Appreciation probability Δ		
	>5%	2.5–5%	0–2.5%	0–2.5%	2.5–5%	>5%
Australia	3.3	4.9	–5.6	–4.4	1.4	0.3
Euro area	0.0	0.9	0.5	–7.7	6.2	0.0
Japan	0.9	0.2	–8.5	–2.5	6.8	3.1
Switzerland	0.0	0.3	–7.8	1.7	6.0	0.0
United Kingdom	0.0	3.1	1.9	–6.4	1.4	0.0
United States	0.0	0.1	–7.6	5.6	1.8	0.0

The table shows changes in appreciation and depreciation probabilities for each currency implied by a shift from the blue to the red distributions shown in Fig. 4, expressed in percentage points

mostly in accord with usual currency characterisations. These results thus complement and qualify the information about local fit from $R^1(\tau)$ measures, and give a much more nuanced depiction of different currencies' tail behaviour.

Such information has many potential applications. For example, in the Online Appendix we attempt a classification of currencies into safe haven-like, risky or generically volatile on that basis (see Sect. 2 of the Online Appendix). Our working definition of safe haven-like behavior applies to currencies that exhibit positive upside entropy and negative downside entropy (so that tail risks shift to the upside as global financial conditions tighten), while risky currencies are defined through the opposite behaviour (so that tail risks shifts to the downside in the same situation); a third category collects currencies that do not fit either pattern. However, a few words of caution are in order. First of all, even 68% confidence bands around our entropy estimates can be wide, so the results need to be treated accordingly. For example, our point estimate for Australia's upside entropy is marginally positive, though statistically insignificant (see Fig. 5) so that in our sample, its currency ends up in the third category rather than in the risky one. Moreover, despite the exercise being done on a NEER basis, the currencies of countries with an exchange rate peg and/or members of a monetary union will tend to display dynamics very much in line with their anchor currencies. This for example explains Peru and Bulgaria's otherwise surprising presence in the safe haven-like category.[17]

3.4 An Application to the COVID-19 Crisis

The sharp currency moves observed amidst the turmoil in global financial markets over March 2020, when the COVID-19 pandemic was sharply accelerating, offer a good setting in which to showcase the advantages of our modelling approach. We do so by feeding our models for the Japanese yen and the Australian dollar from

[17] According to Ilzetzki et al. (2021), Peru had a tight crawling band around the US dollar for a large part of our sample, while Bulgaria had a peg, first to the German mark and then to the euro.

Sect. 3 the observed value of our global financial conditions index for March 2020, and using the resulting conditional distributions to compare the likelihood of the observed currency moves to that implied by average global financial conditions.[18]

Figure 6 reports these distributions, with outturns shown by the black stars. As expected, as global financial conditions tightened sharply, the conditional distributions skewed to the right for the yen and to the left for the Australian dollar. This happened as the yen appreciated by almost 4% and the Australian dollar depreciated by almost 6%. The likelihood of observing such (or larger) moves in FX returns is negligible under average global financial conditions (about 3% and 0.1% for the yen and Australian dollar, respectively) but increases markedly once one takes into account concomitant changes to global financial conditions (to about 30% and 15% for the yen and Australian dollar, respectively).

This exercise shows the usefulness of our model for (i) putting realised currency moves in context by considering changes in global financial conditions and (ii) assigning (conditional) probabilities to every possible FX outturn. In this particular instance, we can see that changes in the value of both the Japanese yen and the Australian dollar were relatively 'extreme' even once the concomitant deterioration in global financial conditions is taken into account, but their likelihood is nevertheless much higher than in a situation in which global financial conditions are closer to their sample mean.

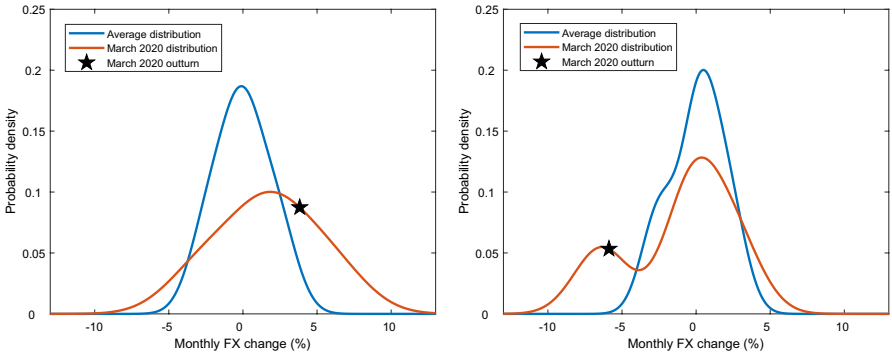
In the next Section we turn to analysing the underlying country characteristics that are associated with such different responses of currencies' distributions to changes in global financial conditions.

4 Identifying Risk Factors: A Portfolio Sorting Approach

What country-level characteristics are associated with the different exchange rate dynamics documented in the previous section? Or, in other words, are there any risk factors associated with specific tail behaviours that policymakers and investors should keep track of? To answer this question, in this Section we draw on portfolio sorting techniques, popular in the FX and equity pricing literatures. In Sect. 4.1 we explain the rationale and mechanics behind our portfolio sorting exercises, while in Sect. 4.2 we identify relevant risk factors by studying the returns, in the face of changes in global financial conditions, of portfolios constructed on their basis.[19] As a tangible output potentially useful for practitioners, we provide a simple tabular mapping between changes in a particular risk factor and increases in currency returns' tail risk.

[18] It should be noted that this is not an out-of-sample exercise, as we are feeding the model *observed* monthly global financial conditions, which are only available at the end of the month just like the left-hand-side variable. Of course, more timely (e.g. weekly or daily) assessments of global financial conditions could be constructed and used to properly nowcast/forecast monthly exchange rate returns.

[19] Throughout the current exercise we do not consider eurozone currencies given the asymmetry between national/domestic risk factors and a zone-wide currency whose value individual countries can only partially influence.



(a) Japan **(b)** Australia

Fig. 6 Conditional distributions and outturns during the COVID-19 outbreak in March 2020. *Note* The panels show, for each currency, nonparametric probability densities fitted to our estimated quantiles conditional on an average (zero) value of global financial conditions (in blue), and on the value of global financial conditions observed in March 2020 (3.7 standard deviations, in red). Overlaid is the realised value of the respective NEERs' returns in March 2020 (black star). (Color figure online)

4.1 Portfolio Sorting

Identifying country characteristics associated with the individual features of exchange rate returns distributions documented in Sect. 3 is challenging: for each country, conditional distributions are estimated from the whole (time series) sample and offer a single summary statistic. However, it is likely that the risk factors associated with such dynamics change over time. For example, it would not be appropriate to try to associate a certain conditional exchange rate distribution to average fiscal deficits over 25 years, as this statistic is likely to hide significant variation over the sample. To address such concerns, we need to introduce a degree of time-variation in our analysis. To do so, we conduct a set of portfolio sorting exercises, widely used in the equity and FX pricing literatures (see Cenedese 2015, and references therein).

We start from a series of candidate variables that have been identified in the literature as being associated with particular reactions of exchange rates to changes in global financial conditions: interest rate differentials (versus the rest of the world), current account balances, fiscal balances, net foreign assets and levels of international reserves.[20] A series of empirical papers have analysed the importance of these risk factors for average exchange rate dynamics: Brunnermeier et al. (2009), Lustig et al. (2011) and Menkhoff et al. (2012) study the risk features of high interest rate currencies, Della-Corte et al. (2016) that of currencies of countries with large external imbalances, while Fratzscher (2009) and Habib and Stracca (2012) assess the relevance of a wide range of variables, including fiscal balances, net foreign assets

[20] Exact definitions and data sources can be found in Appendix A. Interest rate differentials are implied from FX forward contracts. Note that recent CIP deviations mean that there is measurement error in this quantification of interest rate differentials for a portion of our sample.

and international reserves, for currency returns. In turn, these studies are grounded on a rich history of theoretical work linking these macroeconomic variables and exchange rate dynamics.[21,22]

We consider each of the candidate risk factor variables in turn and, at each point in time throughout our sample, begin by ranking countries according to the values they display for the variable under consideration. So, for example, when working with current account balances, we rank countries from those displaying the highest current account surplus to those with the highest deficit.[23]

We then assign currencies to three portfolios according to this ranking. Continuing the previous example, the first portfolio contains the currencies with the largest current account deficits (the 'riskier' set of currencies if current account deficits are indeed a risk factor), while the third portfolio contains the currencies with the largest current account surpluses (the 'safest' group).[24] Finally, we compute the return of each portfolio over the month as the equally weighted return of its component currencies.[25]

The advantage of such a portfolio sorting approach is that it introduces time variation in the exposure to risk factors, which could be associated with particular exchange rate dynamics. This is achieved by allowing countries to have different levels of exposure at different points in time. For example, country A could exhibit a large current account surplus in period t and a large deficit in period $t+k$. In this situation, the return of country A's currency in period t will be assigned to the portfolio comprising surplus countries, while the return in period $t+k$ will be assigned to the portfolio comprising deficit countries. By doing this, our estimates do not depend on the whole time series of returns of a particular country or group of countries, but instead returns are computed dynamically depending on where countries lie in the ranking of risk factors.[26]

We conduct the exercise described above separately for each of the risk factors considered over 1995-2016, and then analyse how exposure to each of them is associated with different tail behaviour in the face of changes in global financial conditions.

[21] See, for example, Krugman (1979), Dornbusch and Fischer (1980), Wijnbergen (1991), Obstfeld and Rogoff (1995) and Gabaix and Maggiori (2015).

[22] Strictly speaking, there is no need for a structural macroeconomic link between the risk factors considered and the exchange rate; as long as investors believe these to be risk factors, and act accordingly, we could still observe the expected conditional correlations, at least in the short run.

[23] We rebalance portfolios annually given the limited availability of data for the sorting variables.

[24] See Sect. 5 for a robustness exercise which considers an alternative version of the portfolio sorting in which the assignment of currencies to risk buckets is performed according to lagged values of the risk variables under consideration.

[25] We use pure FX-driven returns, i.e. log exchange rate changes. The convention matches that of the previous section, so that a positive return corresponds to an appreciation.

[26] In practice, these portfolios are moderately stable but not constant: currencies typically remain in their most common portfolio throughout 68% of the sample, based on an average across risk factors.

4.2 Risk Factors and Global Financial Conditions

After computing portfolio returns as described in Sect. 4.1, we proceed to analyse how their distributions are affected (in sample) by shifts in global financial conditions, as in Sect. 3. In line with the previous section, we first estimate conditional quantile functions for each portfolio returns series, and then fit two empirical distributions: one conditional on average global financial conditions, and another conditional on a one standard deviation tightening of global financial conditions. That is, we want to know the distribution of returns of each portfolio under both 'normal' and 'tight' global financial conditions. If a factor under consideration is indeed a 'true' risk factor, we would expect the return of the riskier portfolio to skew more significantly to the left following a tightening of global financial conditions (that is, we would expect a larger increase in the likelihood of currencies exposed to that factor depreciating sharply in the event of a global shock).

Figure 7 provides an example by reporting distributions of high- and low-risk portfolios (in red and blue, respectively) sorted on the basis of interest rate differentials vis-à-vis the rest of the world, both conditional on average (solid lines) and 'tight' global financial conditions (dashed lines).[27] Two features stand out. First, under average global conditions, the returns of the high-risk portfolio already exhibit 'fatter' tails than those of the low-risk one. Moreover, with tighter global financial conditions, the left tail of the distribution of returns of the high-risk portfolio (in other words, its crash risk) increases sharply, while that of the low-risk portfolio remains largely unchanged. This behaviour suggest that interest rate differentials are indeed a risk factor in the face of tighter global financial conditions, in the sense that they are associated with larger increase in crash risk under such circumstances.

While these directional insights are interesting in their own right, the main advantage of our methodology is that it allows to quantify such tails risks. We therefore estimate conditional distributions for the returns of portfolios sorted on the basis of all risk factors we consider, and use them to assess the probabilities of a range of possible outcomes under different scenarios for global financial conditions. A tangible output of this exercise is a simple tabular mapping between changes in exposures to the candidate risk factors and the implied changes in tail risks following a tightening of global financial conditions within our sample.[28]

Table 5 reports our results. The first observation is that under average global financial conditions, riskier portfolios—as sorted by the candidate risk factors—display negligible crash risk (defined as the probability of losses exceeding 2.5%) in all cases (col. IV). However, this changes once we condition on tighter global financial conditions. In such a situation, high-risk portfolios sorted on the basis of interest rate differentials, level of international reserves and fiscal balances display significantly higher crash risk than their low-risk alternatives. In contrast, crash risk

[27] Analogous figures for the remaining risk factors can be found in Sect. 3 of the Online Appendix.

[28] In particular, we consider global financial conditions three standard deviations above their average level. This compares with prints of 6.6 in October 2008 during the global financial crisis, and 3.7 in March 2020, at the height of COVID-19-related stress.

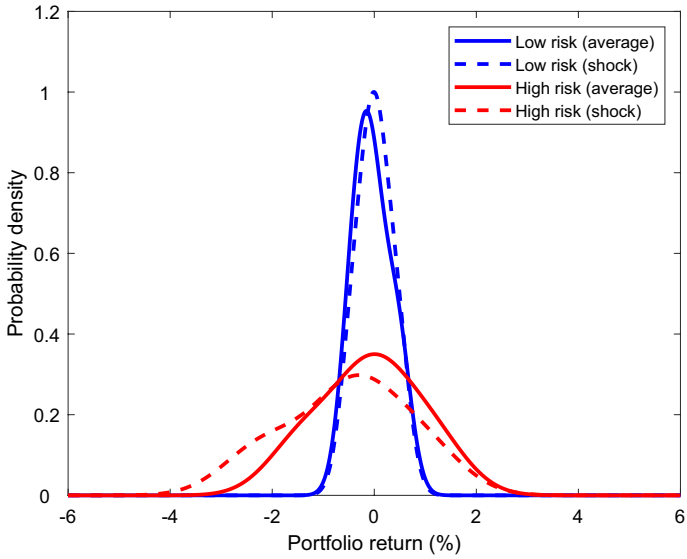


Fig. 7 Return distributions of portfolios sorted by interest rate differentials. *Note* The figure shows non-parametric probability densities estimated for the returns of low- (in blue) and high- risk portfolios (in red) sorted on the basis of interest rate differentials under average global financial conditions ('average', solid lines) and global financial conditions one standard deviation tighter than average ('shock', dashed lines). (Color figure online)

increases by similar amounts for high- and low-risk portfolios sorted on the basis of current account balances or net foreign asset positions. Thus overall, only a subset of the risk factors is positively correlated with 'crash risk' in the face of a tightening in global financial conditions.

This exercise also allows to construct an in-sample mapping from changes in risk factors to exchange rate crash risk in the event of tighter global financial conditions. For example, a country whose interest rate differential with respect to the rest of the world increases by 8.2 p.p. (therefore moving from the middle-risk portfolio to the high-risk one based on averages over our sample, see col. III) should see an increase in the probability of its currency depreciating by more than 2.5% of approximately 25 p.p. in the event of a sharp tightening of global financial conditions.[29]

These findings are consistent with (yet more nuanced than) the mean-based results in Brunnermeier et al. (2009), Lustig et al. (2011), Menkhoff et al. (2012), Della-Corte et al. (2016), Fratzscher (2009) and Habib and Stracca (2012). More precisely, our results show that currencies of countries displaying large values for

[29] Strictly speaking, this increase in interest rate differential delivers, on average over the sample, a move from the average position in the middle-risk portfolio to the average position in the high-risk portfolio, while of course a smaller change might already lead to a portfolio change with the consequent increase in crash risk (e.g. if starting from a higher differential within the medium-risk portfolio and moving to the lower end of the high-risk portfolio).

Table 5 Crash risk probabilities of portfolios under two scenarios for financial conditions

(I) Risk factor	(II) Portfolio	(III) Av. risk factor (%)	(IV) Crash risk (average, %)	(V) Crash risk (shock, %)
Interest rates	H	6.5	1.2	26.6
	M	–1.7	0.0	0.2
	L	–4.3	0.0	0.0
Reserves	H	3.5	2.6	25.5
	M	8.4	0.0	9.0
	L	21.1	0.0	0.0
Fiscal	H	–5.3	0.0	9.2
	M	–2.0	0.0	5.2
	L	2.4	0.0	1.6
CA	H	–4.3	0.0	13.2
	M	–0.6	0.0	21.8
	L	5.1	0.0	13.6
NFA	H	–64.1	0.4	22.8
	M	–17.1	0.0	14.7
	L	54.5	0.0	25.0

The table shows the probabilities of returns losses greater than 2.5% ('crash risk') for high-, medium- and low-risk portfolios as sorted by the range of risk factors listed in col. I, both under average global financial conditions (col. IV) and global financial conditions three standard deviations tighter than average (col. V). The average value of the risk factor for the currencies comprising each portfolio is also reported (col. III). The table allows to compute, for each risk factor, the (average) required change in the risk factor for a currency to switch between the H, M and L portfolios, and the implied change in crash risk under alternative scenarios for financial conditions

the risk factors considered not only experience a reduction in their conditional mean in the face of tighter global conditions, but the resulting distributions change shape, with a more than proportional increase in the probability mass assigned to the left (depreciation) tails for some of the risk factors.

In sum, in this section we showed that a number of intuitive risk factors contain useful information for quantifying exchange rate risk in the face of tighter global financial conditions. These insights should be of interest to policymakers assessing the financial stability outlook of countries, and to investors performing risk management calculations for their investment strategies, which naturally rely on tail risk information.

5 Robustness

In this section we list a series of robustness checks on our baseline results along four dimensions. First, we show that richer specifications of the conditional quantile function do not alter our key results. Second, we study alternative exchange rate measures, namely US dollar bilaterals and currencies' excess returns (net of interest

rate differentials). Third, we modify our exercise in Sect. 4 by sorting portfolios using lagged values of the sorting variables to rule out reverse causality issues. And finally, we consider a range of alternative proxy measures for global financial conditions. All results are available in Sect. 4 of the Online Appendix.

5.1 Specification of the Quantile Function

Our baseline specification, shown in Eq. 1, is deliberately parsimonious. In this section we explore the consequences of more complex specifications along three dimensions: a generic control for omitted variables by means of a lagged FX returns term; a control for 'standard' factors typically found to be relevant for FX returns; and a control for possible nonlinear effects *in the quantile function itself*.

A straightforward way of checking whether the specification suffers from an obvious omitted variables problem is to add lagged exchange rate returns to the explanatory variables. The resulting specification then becomes a quantile autoregression (Koenker et al. 2006), with an added exogenous regressor. The resulting estimates of $\beta_h(\tau)$ (that is, the coefficient corresponding to our global financial conditions index) are usually very similar to the baseline. Charts can be found in Sect. 4.1.1 of the Online Appendix.

In a separate exercise, we augment our specification by adding 'standard' factors used to account for the factor structure usually found in exchange rate returns. Specifically, we include a 'dollar' factor, which is the average return across portfolios of interest rate-sorted US dollar bilateral exchange rates, and a 'carry' factor, which computes the return of going 'long' on the portfolio of currencies with the highest interest rate differentials (vis-à-vis the US), and 'short' on the portfolio of currencies with the lowest interest rate differentials. Data for the returns of both factors comes from Lustig et al. (2011). We find that while these two additional factors tend to be significant in explaining the returns of the currencies in our sample (as expected), they do typically leave the coefficient corresponding to our global financial conditions index broadly unchanged, which suggests that they contain different information. Charts can be found in Sect. 4.1.2 of the Online Appendix.

Finally, we also explore the potential role of nonlinearities in the quantile function itself by considering an additional quadratic term for our global financial conditions index.[30] We find that the coefficient corresponding to this additional term is usually small and typically non-significant for most of the currencies in our sample. Moreover, the coefficients are also typically constant across quantiles, which therefore results in small parallel shifts in the coefficient on the linear term, which nevertheless maintains its 'shape' across quantiles (and its significance). Charts can be found in Sect. 4.1.3 of the Online Appendix.

[30] As opposed to the nonlinear relationship between the dependent variable and the regressors that is already inherent in the quantile regression approach.

5.2 Alternative Measures of Exchange Rate Returns

Our baseline results in Sect. 3 are based on nominal effective exchange rates (NEERs). This choice is motivated by the desire to focus on plain exchange rate moves, abstracting from interest rate differentials, and to avoid US-driven, globally synchronised changes in bilateral dollar exchange rates. However, in order to facilitate comparisons with the existing literature, we also report results of exercises that consider alternative choices: NEERs-based excess returns and US dollar bilaterals. Charts showing downside and upside entropies for comparison with our baseline results can be found in Sect. 4.2 of the Online Appendix.

Results are broadly unchanged when considering excess returns, which net out interest rate differentials of the currency under consideration vis-à-vis the rest of the world. For US dollar bilaterals the changes are also small. Tail behaviour rankings based on relative entropy are virtually unaltered, despite changes in their values in the expected direction: given the high conditional upside entropy of the US dollar NEER itself, the conditional upside entropies of other safe haven currencies become smaller when considering dollar bilaterals, while the downside entropies of risky currencies increase.

5.3 Portfolio Sorting Strategy

In our baseline results, the sorting of currencies into portfolios based on the values of risk factors is done based on contemporaneous values of the risk factors. More specifically, the sorting is done at annual frequency due to data availability, while the conditional returns of the resulting portfolios are measured at monthly frequency. One downside of such strategy is that it is liable to suffering from reverse causality, because the risk factors could themselves change in response to changes in global financial conditions over the year, in turn affecting the composition of portfolios. A solution is to perform the portfolio sorting using lagged values of the risk factors, which comes at the cost of potentially using out-of-date data, given the annual rebalancing of portfolios but monthly returns computation. With this caveat in mind, we check the robustness of the results reported in Sect. 4 to this alternative sorting strategy, and find that our results, reported in Sect. 4.3 of the Online Appendix, are broadly unchanged.

5.4 Measurement of Global Financial Conditions

Our global financial conditions index is one of many attempts in the literature to summarise moves in global risky asset prices. Other available measures put forward include the global factor presented in Miranda-Agrippino and Rey (2015) (MAR), the RORO index introduced by Chari et al. (2020) and the index introduced by Bekaert et al. (forthcoming) (BEX). Moreover, the VIX index, despite being specific to the US stock market, has been used as yet another gauge of

global risk (see, for example, Habib and Stracca (2012), who identify their global risk shock based on VIX data).

In this section we compare the goodness of fit achieved by the global indices listed above when trying to fit variation in the tail regions of the distributions of returns of the currencies in our sample (as in Table 3 for our baseline results). To streamline the exposition, we focus on the average R^1 across tail quantiles achieved by each index for the tail region that it fits best for each currency considered.[31] We first compare specifications which consider each index in turn, benchmarking them against our baseline specification (with our GFCI index alone) over a common sample. We subsequently consider a specification which includes several indices jointly.[32]

The pairwise comparisons in columns (1) to (8) of Table 6 show that, apart from the Miranda-Agrippino and Rey (2015) index, which seems to have a somewhat lower explanatory power for FX returns, the rest of the measures deliver broadly similar results. Our GFCI index typically delivers higher R^1 measures than the RORO and BEX indices, while the comparison with the VIX index (cols. 1-2) is more nuanced, as the VIX delivers a higher R^1 for the aggregate sample, but performs worse when the comparison is restricted to EMs. The latter result is not surprising given the global nature of our index and the US-specific nature of the VIX.

The specification that adds the VIX, RORO and BEX indices jointly with our GFCI (col. 10) fares better than a GFCI-only alternative over the same sample (col. 9), which suggest that, despite sharing a strong common component (as seen in the high correlations documented in Table 2 and also suggested by the first eight columns of Table 6), there is a degree of complementary information contained in them that can be exploited for fitting exchange rate tail risk even better.

6 Conclusion

We provide novel empirical evidence on the relationship between the entire distribution of currency returns and global financial conditions. Our results corroborate some of the prevailing narratives about safe haven and risky currencies, but also provide richer insights than existing studies focussing on mean returns, allowing for example to rank currencies according to their tail behaviour and to quantify the shifts in their distributions following changes in global financial conditions. We also document the role of commonly used macro-financial risk factors in explaining losses on FX portfolios in the face of tighter global financial conditions. These insights should be of interest to policymakers assessing the financial stability outlook of countries, and to investors performing risk management calculations for their investment strategies.

[31] Specifically, we report the highest of the two average R^1s between percentiles 5 to 25 and 75 to 95, respectively.

[32] We exclude the MAR index for this exercise given that it ends early in our sample and would therefore have resulted in a significant loss of observations.

Table 6 Goodness of fit measure for 'best' tail

	VIX		MAR		RORO		BEX		ALL	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Australia	7.8	6.0	10.0	0.4	8.2	5.8	7.8	6.4	8.2	15.3
Japan	8.6	15.1	8.3	0.7	14.5	5.0	8.6	10.0	14.5	22.4
Euro area	4.0	4.3	3.4	0.5	1.1	2.4	4.0	3.8	1.1	3.9
Switzerland	5.8	6.3	6.7	0.6	5.3	0.2	5.8	3.6	5.3	10.9
UK	3.4	4.2	6.4	0.9	5.2	9.5	3.4	4.7	5.2	10.1
US	4.5	5.2	9.5	1.2	6.7	2.5	4.5	5.4	6.7	11.8
Full sample	4.0	4.4	4.5	1.6	4.6	3.1	4.0	3.7	4.6	8.2
Liquid EMs	5.2	5.1	6.4	1.8	6.9	2.9	5.2	5.0	6.9	10.7
BRICS	5.2	4.3	6.7	2.4	6.5	3.2	5.2	4.5	6.5	10.7

The table shows goodness of fit measures for the most accurately fitted tail in each case. This is computed as the highest of the average $R^1(\tau)$ between the 5th and 25th percentiles for the left tail, and between the 75th and 95th percentiles for the right tail. 'Full sample' refers to the average of this measure across the 61 currencies in our sample, 'Liquid EMs' to the average measure for the currencies of China, Hong Kong, Korea, Singapore, Mexico, India, Russia, South Africa, Turkey and Brazil (which were the ten most traded EM currencies in 2019 according to Schrimpf and Sushko 2019) and 'BRICS' to the average measure for the currencies of Brazil, Russia, India, China and South Africa. Each block of two columns compares a specification which includes our GFCI alone first, and a specification that contains an alternative index alone in the second place, for the same sample. GFCI refers to our Global Financial Conditions Index, VIX to the VIX index, MAR to the index from Miranda-Agrippino and Rey (2015), RORO to the index from Chari et al. (2020) and BEX to the index in Bekaert et al. (forthcoming). ALL refers to a specification that includes the VIX, the RORO and BEX indices, as well as our GFCI

Appendix: Data

Definitions and Sources

See Table 7 for definition of variables and data sources.

Table 7 Variable definitions and data sources

Variable	Definition	Source
$\Delta FX_{i,t}$	Monthly change in the broad nominal effective exchange rate.	BIS
$GFCI_t$	Global financial conditions index (see below for underlying components)	Authors' calculation
GFCI components		
Long-term gov. interest rate	Yield on nominal government bonds with maturity of 10 years.	Thomson Reuters Datastream
Sovereign spreads	Advanced economies: difference between domestic long-term government interest rates and those of bonds of a benchmark country (Germany for Europe and US for rest of the world).	Thomson Reuters Datastream
	Emerging market economies: stripped spreads from JP Morgan's EMBI.	JP Morgan
Term spreads	Difference between domestic long-term government interest rates and a domestic short term T-bill rate (with maturity of 3 months or closest).	Thomson Reuters Datastream and Bank of America Merrill Lynch.
Interbank spreads	Difference between 3-month interbank rate (or closest) and 3-month T-Bill rate (or closest).	Thomson Reuters Datastream and national central banks.
Corporate spreads	Corporate spread indices.	Bank of America Merril Lynch, Barclays, JPMorgan (CEMBI) and Standard & Poor's.
Equity returns	3-month return of domestic stock index, measured in domestic currency.	Thomson Reuters Datastream
Equity volatility	Realised monthly volatility computed using daily changes in equity index.	Thomson Reuters Datastream
Market capitalisation of financial sector	Market capitalisation of MSCI Country Financials Index divided by MSCI Country Index.	MSCI Inc.
Risk factors		
Forward discount	Forward discount of currency relative to the rest of the currencies in our sample. Bilateral forward discounts obtained as the percentage difference between the spot exchange rate and a 3-month exchange rate forward contract.	Thomson Reuters Datastream
Reserves	Total international reserves of country as a share of GDP.	IMF IFS

Table 7 (continued)

Variable	Definition	Source
Fiscal balance	Fiscal position of the government after accounting for capital expenditures as a share of GDP.	OECD
Current account	Current account balance as a share of GDP.	IMF IFS and OECD
Net foreign assets	Net international investment position (claims on non residents minus liabilities to them).	IMF IIP/BOP interpolated with Lane and Milesi-Ferretti (2018) when missing

Samples

Exchange rates. The analysis in Sect. 3 is conducted using Nominal Effective Exchange Rates (NEERs) from the BIS from January 1994 to June 2018 for the following countries: Algeria, Argentina, Australia, Austria, Belgium, Brazil, Bulgaria, Canada, Chile, China, Chinese Taipei, Colombia, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Euro area, Finland, France, Germany, Greece, Hong Kong, Hungary, Iceland, India, Indonesia, Ireland, Israel, Italy, Japan, Korea, Latvia, Lithuania, Luxembourg, Malaysia, Malta, Mexico, Netherlands, New Zealand, Norway, Peru, Philippines, Poland, Portugal, Romania, Russia, Saudi Arabia, Singapore, Slovakia, Slovenia, South Africa, Spain, Sweden, Switzerland, Thailand, Turkey, United Arab Emirates, United Kingdom, United States and Venezuela.

Exchange rate changes are computed as log differences on monthly averages; interest rate differentials (and the corresponding excess returns) are not considered in the baseline analysis.

Financial condition indices. We compute FCIs at the monthly frequency from April 1995 to June 2018 for the following countries: Argentina, Australia, Austria, Belgium, Brazil, Bulgaria, Canada, Chile, China, Colombia, Czech Republic, Denmark, Finland, France, Germany, Hungary, India, Ireland, Israel, Italy, Japan, Korea, Malaysia, Mexico, Netherlands, New Zealand, Norway, Peru, Philippines, Poland, Portugal, Russia, South Africa, Spain, Sweden, Switzerland, Thailand, Turkey, United Kingdom, United States and Venezuela.

Armed with a set of country-specific FCIs we then compute our proxy of global financial conditions as the first principal component of these.

Appendix: Quantile Regression

Given a linear model for the conditional quantile function

$$Q_y(\tau|X) = x\beta(\tau) \tag{B.1}$$

the quantile regression estimate $\hat{\beta}(\tau)$ is the minimiser of

$$\hat{V}(\tau) = \min_{\beta\in\mathbb{R}^p} \sum \rho_\tau\left(y_i - x_i'\beta\right) \tag{B.2}$$

where $\rho_\tau(u) = u[\tau - I(u < 0)]$ is the so-called check function.

As discussed in Koenker (2005), the solution of problem B.2 is amenable to linear programming techniques. However, in our MATLAB implementation, we have found it computationally more efficient to approximate the exact solution via an iteratively-reweighted-least-squares (IRLS) algorithm. This is motivated by the close relationship of B.2 to the problem of finding the least-absolute-deviations (LAD) estimator (which obtains for $\tau = 0.5$), and more generally of solving L^p–norm linear regression problems. Building on Mohammadi (2009), we proceed as follows: we start from an initial OLS estimate,

$$\hat{\beta}^{(0)}(\tau) = \left(x'x\right)^{-1} x'y.$$

We then take the residuals $\hat{u}_i^{(0)}(\tau) = y_i - x_i\hat{\beta}^{(0)}(\tau)$ and construct a diagonal matrix of weights $w^{(t)}, t > 0$, whose diagonal elements are given by

$$w_{ii}^{(t)}(\tau) = \frac{1}{\rho_{1-\tau}\left(u_i^{(t-1)}(\tau)\right)}$$

We then obtain an updated estimate $\hat{\beta}^{(t)}(\tau)$, residuals $\hat{u}^{(t)}(\tau)$ and weights $w^{(t+1)}(\tau)$ using weighted least squares:

$$\hat{\beta}^{(t)}(\tau) = \left(x'w^{(t)}(\tau)'x\right)^{-1} x'w^{(t)}(\tau)'y$$

and iterate until convergence. Essentially, the procedure approximates B.2 by a convergent sequence of weighted sums of square residuals, where the weights are chosen so as to approximate the check function ρ_τ with a quadratic one.

Bootstrapping

While there are several results available for inference in quantile regression with time-series data (see for example Xiao (2012), Zhou and Shao (2013)), we take a shortcut and deal with potential autocorrelation in the errors from B.2 by bootstrapping confidence intervals for all quantities of interest. Fitzenberger (1998) shows that a moving (or overlapping) block bootstrap procedure provides heteroskedasticity- and autocorrelation-consistent (HAC) standard errors for quantile regression coefficient estimators.

The procedure works as follows: letting $z_t = [y_t, x_t]$ denote the original data, T the sample size and b a suitably chosen block length, a resample z_{it}^* of length $T^* = b * round(T/b)$ is obtained by joining $round(T/b)$ draws (with replacement) of b consecutive elements of z_t (blocks), where the blocks are allowed to overlap. Each resample z_{it}^* yields an estimate of the quantile regression coefficients $\hat{\beta}_i^*(\tau)$ and can be used to compute all other statistics of interest, such as $\hat{V}_i(\tau)$ and thus $R^1(\tau)$ etc. Confidence intervals at level γ for $\hat{\beta}(\tau)$ and other quantities of interest are computed as

$$\left(2\hat{\beta}(\tau) - \hat{\beta}^*_{\frac{1-\gamma}{2}}(\tau), 2\hat{\beta}(\tau) - \hat{\beta}^*_{\frac{\gamma}{2}}(\tau)\right) \tag{B.3}$$

where $\hat{\beta}_p^*(\tau)$ denotes the $p-$th percentile of the bootstrapped draws $\hat{\beta}_i^*(\tau)$[33].

[33] In the computation of confidence intervals for $R^1(\tau)$ we instead compute directly percentiles from the bootstrapped draws to ensure non-negative values.

Fitting a Nonparametric Distribution to the Quantiles

We seek to match the fitted quantiles implied by the model in Eq. 1, $\hat{q}(\tau)$, conditional on average global financial conditions (which are 0 by construction), given simply by $\hat{q}_{GFC=0}(\tau) = \hat{\alpha}_h(\tau)$, and conditional on a one standard deviation increase in global financial conditions, given by $\hat{q}_{GFC=1}(\tau) = \hat{\alpha}_h(\tau) + \hat{\beta}_h(\tau)$, with the quantiles of nonparametric distributions with Normal kernel ϕ and suitably chosen bandwidth h, whose generic density function is given by:

$$f_h(x) = \frac{1}{Ih} \sum_{i=1}^{I} \phi\Big(\frac{x - z_i}{h}\Big). \tag{B.4}$$

This is accomplished by solving the following minimisation problem for the two sets of fitted quantiles $\hat{q}(\tau)$:

$$\min_{\{z_i\}_1^I} \sum_{\tau} \left[\hat{q}(\tau) - Q_h(\tau|\{z_i\}_1^I)\right]^2 \tag{B.5}$$

where $Q_h(\tau|\{z_i\}_1^I)$ is the τ-th quantile function of $f_h(x)$, which depends on a set of I artificial observations z_i that are chosen so as to match the fitted quantiles.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

Adrian, T., N. Boyarchenko, and D. Giannone. 2019. Vulnerable growth. *American Economic Review* 109: 1263–1289.

Adrian, T., Grinberg, F., Liang, N., and Malik, S. 2018. The term structure of growth-at-risk. IMF Working Papers 18/180, International Monetary Fund.

Arregui, N., Elekdag, S., Gelos, R.G., Lafarguette, R., and Seneviratne, D. 2018. Can countries manage their financial conditions amid globalization?. IMF Working Papers 18/15, International Monetary Fund.

Arrigoni, S., Bobasu, A., Venditti, F. 2020. The simpler the better: measuring financial conditions for monetary policy and financial stability. Working Paper Series 2451, European Central Bank.

Avdjiev, S., V. Bruno, C. Koch, and H.S. Shin. 2019. The dollar exchange rate as a global risk factor: Evidence from investment. *IMF Economic Review* 67: 151–173.

Bekaert, G., Engstrom, E.C., and Xu, N.R. forthcoming. The time variation in risk appetite and uncertainty. *Management Science*.

Brunnermeier, M.K. 2009. Deciphering the liquidity and credit crunch 2007–2008. *Journal of Economic Perspectives* 23: 77–100.

Brunnermeier, M.K., S. Nagel, and L.H. Pedersen. 2009. Carry Trades and Currency Crashes. In *NBER Macroeconomics Annual 2008, Volume 23*, 313–347. National Bureau of Economic Research, Inc, NBER Chapters.

Bruno, V., and H.S. Shin. 2014. Cross-border banking and global liquidity. *The Review of Economic Studies* 82: 535–564.

Cenedese, G. 2015. Safe haven currencies: A portfolio perspective. Bank of England working papers 533, Bank of England.

Cenedese, G., L. Sarno, and I. Tsiakas. 2014. Foreign exchange risk and the predictability of carry trade returns. *Journal of Banking and Finance* 42: 302–313.

Cerutti, E., Claessens, S., and Rose, A.K. 2017. How important is the global financial cycle? Evidence from capital flows. NBER Working Papers 23699, National Bureau of Economic Research, Inc.

Cesa-Bianchi, A., A. Ferrero, and A. Rebucci. 2018. International credit supply shocks. *Journal of International Economics* 112: 219–237.

Cesa-Bianchi, A., Pesaran, M.H., and Rebucci, A. 2018b. Uncertainty and economic activity: A multi-country perspective. Working Paper 24325, National Bureau of Economic Research.

Chari, A., Stedman, K.D., and Lundblad, C. 2020. Capital flows in risky times: Risk-on/risk-off and emerging market tail risk. NBER Working Papers 27927, National Bureau of Economic Research, Inc.

Crump, R.K., Giannone, D., and Hundtofte, S. 2018. Changing risk-return profiles. Staff Reports 850, Federal Reserve Bank of New York.

Della-Corte, P., S.J. Riddiough, and L. Sarno. 2016. Currency premia and global imbalances. *Review of Financial Studies* 29: 2161–2193.

Dornbusch, R., and S. Fischer. 1980. Exchange rates and the current account. *American Economic Review* 70: 960–971.

Drehmann, M., Borio, C., and Tsatsaronis, K. 2012. Characterising the financial cycle: don't lose sight of the medium term!. BIS Working Papers 380, Bank for International Settlements.

Eguren-Martin, F., O'Neill, C., Sokol, A., von dem Berge, L. 2020. Capital flows-at-risk: push, pull and the role of policy. Bank of England working papers 881, Bank of England.

Farhi, E., Fraiberger, S.P., Gabaix, X., Ranciere, R., and Verdelhan, A. 2009. Crash Risk in Currency Markets. NBER Working Papers 15062, National Bureau of Economic Research, Inc.

Farhi, E., and X. Gabaix. 2016. Rare disasters and exchange rates. *The Quarterly Journal of Economics* 131: 1–52.

Fitzenberger, B. 1998. The moving blocks bootstrap and robust inference for linear least squares and quantile regressions. *Journal of Econometrics* 82: 235–287.

Forbes, K.J., and F.E. Warnock. 2012. Capital flow waves: Surges, stops, flight, and retrenchment. *Journal of International Economics* 88: 235–251.

Fratzscher, M. 2009. What explains global exchange rate movements during the financial crisis? *Journal of International Money and Finance* 28: 1390–1407.

Gabaix, X., and M. Maggiori. 2015. International liquidity and exchange rate dynamics. *The Quarterly Journal of Economics* 130: 1369–1420.

Gaglianone, W.P., and L.R. Lima. 2012. Constructing density forecasts from quantile regressions. *Journal of Money, Credit and Banking* 44: 1589–1607.

Gelos, R.G., Gornicka, L., Koepke, R., Sahay, R., and Sgherri, S. 2019. Capital flows at risk: Taming the Ebbs and flows. IMF Working Papers 19/279, International Monetary Fund.

Ha, J., Kose, M.A., Otrok, C., and Prasad, E.S. 2018. Global Macro-Financial Cycles and Spillovers. Unpublished manuscript.

Habib, M.M., and L. Stracca. 2012. Getting beyond carry trade: What makes a safe haven currency? *Journal of International Economics* 87: 50–64.

Ilzetzki, E., Reinhart, C.M., and Rogoff, K.S. 2021. Rethinking Exchange Rate Regimes. NBER Working Papers 29347, National Bureau of Economic Research, Inc.

Koenker, R. 2005. Quantile regression. In *Econometric society monographs*. Cambridge: Cambridge University Press.

Koenker, R., and G. Bassett. 1978. Regression quantiles. *Econometrica* 46: 33–50.
Koenker, R., and J.A.F. Machado. 1999. Goodness of fit and related inference processes for quantile regression. *Journal of the American Statistical Association* 94: 1296–1310.
Koenker, R., Z. Xiao, J. Fan, Y. Fan, M. Knight, M. Hallin, B.J.M. Werker, C.M. Hafner, O.B. Linton, and P.M. Robinson. 2006. Quantile autoregression [with comments, rejoinder]. *Journal of the American Statistical Association* 101: 980–1006.
Koop, G., and D. Korobilis. 2014. A new index of financial conditions. *European Economic Review* 71: 101–116.
Korobilis, D. 2017. Quantile regression forecasts of inflation under model uncertainty. *International Journal of Forecasting* 33: 11–20.
Krugman, P. 1979. A model of balance-of-payments crises. *Journal of Money, Credit and Banking* 11: 311–325.
Lane, P.R., and G.M. Milesi-Ferretti. 2018. The external wealth of nations revisited: International financial integration in the aftermath of the global financial crisis. *IMF Economic Review* 66: 189–222.
Lustig, H., N. Roussanov, and A. Verdelhan. 2011. Common risk factors in currency markets. *Review of Financial Studies* 24: 3731–3777.
Menkhoff, L., L. Sarno, M. Schmeling, and A. Schrimpf. 2012. Carry trades and global foreign exchange volatility. *Journal of Finance* 67: 681–718.
Miranda-Agrippino, S., and Rey, H. 2015. US Monetary Policy and the Global Financial Cycle. NBER Working Papers 21722, National Bureau of Economic Research, Inc.
Mohammadi, S. 2009. QUANTILEREG: MATLAB function to estimate quantile regression. Statistical Software Components. Boston College Department of Economics.
Obstfeld, M., and K. Rogoff. 1995. Exchange rate dynamics redux. *Journal of Political Economy* 103: 624–660.
Ranaldo, A., and P. Soderlind. 2010. Safe haven currencies. *Review of Finance* 14: 385–407.
Rey, H. 2013. Dilemma not trilemma: The global cycle and monetary policy independence. *Proceedings—Economic Policy Symposium—Jackson Hole*, 1–2.
Rostagno, M., C. Altavilla, G. Carboni, W. Lemke, R. Motto, A.S. Guilhem, and J. Yiangou. 2021. *Monetary policy in times of crisis: A tale of two decades of the European Central Bank*. Oxford: Oxford University Press.
Schrimpf, A., and Sushko, V. 2019. Sizing up global foreign exchange markets. *BIS Quarterly Review*.
Wijnbergen, S.V. 1991. Fiscal deficits, exchange rate crises and inflation. *Review of Economic Studies* 58: 81–92.
Xiao, Z. 2012. 9 - Time Series Quantile Regressions. In *Time Series Analysis: Methods and Applications*, ed. T.S. Rao, S.S. Rao, and C. Rao, Elsevier, vol. 30 of *Handbook of Statistics*, 213–257.
Zhou, Z., and X. Shao. 2013. Inference for linear models with dependent errors. *Journal of the Royal Statistical Society Series B* 75: 323–343.

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